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Course: DLBCSEBI02 – Project: Business Intelligence

Project Report

Task 2: The Application of the Data Vault Method Using the Example of a Receipt

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1. INTRODUCTION

The need for robust data management and analytics infrastructure has grown with the rapid development of digital business environments. Today, organizations generate large volumes of transactional data, and the ability to store, integrate, and analyze this data effectively has become a vital competitive edge. Business intelligence (BI) solutions form the foundation of data-driven decision-making by enabling organizations to turn raw operational data into actionable insights.

The data warehouse (DWH), which acts as a central repository for structured data collected from numerous source systems, is vital to any business intelligence system. The scalability, maintainability, and analytical performance of a data warehouse are heavily impacted by its architecture and modeling. The Data Vault technique and the star schema are two fundamentally different modeling approaches.

Ralph Kimball developed the star schema, which provides a clear, denormalized, query-optimized structure well-suited to static reporting contexts. Dan Linstedt's Data Vault approach, by contrast, offers a highly standardized, audit-friendly, and flexible architecture intended for settings in which needs, business rules, and data sources are constantly changing.

Using a real-world example, a retail purchase receipt, this project report compares these two modeling techniques. A clear example of both approaches is the receipt, which includes details on purchased goods, quantities, prices, transaction date, and store location.

The creation of data models using both the star schema and Data Vault methods is preceded by a detailed analysis of the Data Vault approach at the start of the report. The report's conclusion includes a critical evaluation of the advantages and disadvantages of each strategy.

2. ANALYSIS AND RESEARCH OF THE DATA VAULT METHOD

2.1 Origin and Definition

Dan Linstedt developed the Data Vault technique in the early 1990s, and it was officially introduced to the data warehousing community in the early 2000s. Linstedt developed the technique in response to the limitations of earlier methods, including their inadequate handling of historical data and their inability to adapt quickly to changing business needs (Linstedt & Olschimke, 2015). The term "Data Vault" was coined to describe a long-term, secure repository for company data, similar to a real vault that safeguards priceless objects without altering them.

Typically, a data vault consists of normalized tables that support one or more business functions. These tables are detailed, tracked over time, and uniquely connected. To combine the flexibility of 3NF with the query performance of dimensional models, a hybrid approach merges ideas from both dimensional modeling (star schema) and third normal form (3NF) (Hultgren, 2012).

Three core design principles underpin the methodology**:** complete auditability, support for parallel loading, and alignment with business needs. Data Vault is designed for agile development environments in which source systems frequently change, and new data must be added incrementally, unlike the star schema, which relies on a relatively stable source environment (Collier, 2011).

2.2 Main Components of Data Vault

Hubs, Links, and Satellites are the three primary components specified in the original Data Vault 1.0 standard. Each has a specific architectural function.

2.2.1 Hubs

The core components of the Data Vault model are hubs. For each specific business concept, such as a product, customer, or transaction, each Hub represents a unique set of business keys. A hub only includes the business key, load date, surrogate key (hash key), and record

source. Importantly, hubs do not contain descriptive attributes. Their main role is to act as a stable reference point for the model and to document the existence of a business entity. Hubs remain unchanged even when source systems evolve because they are based on business keys instead of surrogate keys from source systems (Linstedt & Olschimke, 2015).

2.2.2 Links

Links represent the connections and linkages between two or more Hubs. A Link table documents the business transactions or events that connect multiple entities, such as a purchase event that links a product, a transaction, a time, and a location. Links include a load date, a record source, and hash keys for all connected Hubs. They do not contain descriptive information about the connection itself. The model can easily depict many-to-many relationships without data duplication thanks to fully normalized links (Hultgren, 2012).

2.2.3 Satellites

All descriptive, contextual, and historical data on the entities or relationships specified in Hubs and Links are stored in satellites. Multiple Satellite tables, each focused on a specific context or data source, can exist within each Hub or Link. Satellites include the parent hash key, load date, finish date (for historization), hash difference column (for change detection), and descriptive attributes. The Data Vault is inherently capable of serving as a comprehensive audit trail because satellites enable it to track changes over time through an integrated historization process (Linstedt & Olschimke, 2015).

2.3 Data Vault 2.0

In 2015, Dan Linstedt and Michael Olschimke released Data Vault 2.0, which expanded the original standard to meet contemporary business needs. Big Data processing (Hadoop, NoSQL), the use of hash keys rather than sequences for surrogate key generation, support for soft business rules in the Information Mart layer, and methodological alignment with Agile, CMMI, and Six Sigma practices are some of the major improvements in version 2.0 (Linstedt & Olschimke, 2015).

The Staging Area, the Raw Data Vault (the fundamental Hub, Link, and Satellite model), and the Business Vault or Information Mart layer, where business rules are implemented and data is ready for consumption; are the three levels that make up the architecture of Data Vault 2.0. One of the methodology's biggest benefits is the tiered separation of raw data from business logic, which preserves raw historical data as business rules change (Collier, 2011).

Furthermore, Data Vault 2.0 adds two new table types: Point-in-Time (PIT) tables and Bridge tables, which are performance optimization structures built in the Business Vault layer to enable effective querying across multiple Satellites, and Reference Tables, which store static or semi-static lookup data.

3. STAR SCHEMA DATA MODEL

3.1 Introduction to the Star Schema

One of the most popular data modelling techniques for OLAP (Online Analytical Processing) and data warehousing systems is the star schema. The star schema, which was first codified by Ralph Kimball in the 1990s, arranges data into a core fact table encircled by many dimension tables, giving the structure a star-like appearance (Corr & Stagnitto, 2011). The method puts analytical performance and query ease ahead of storage normalization.

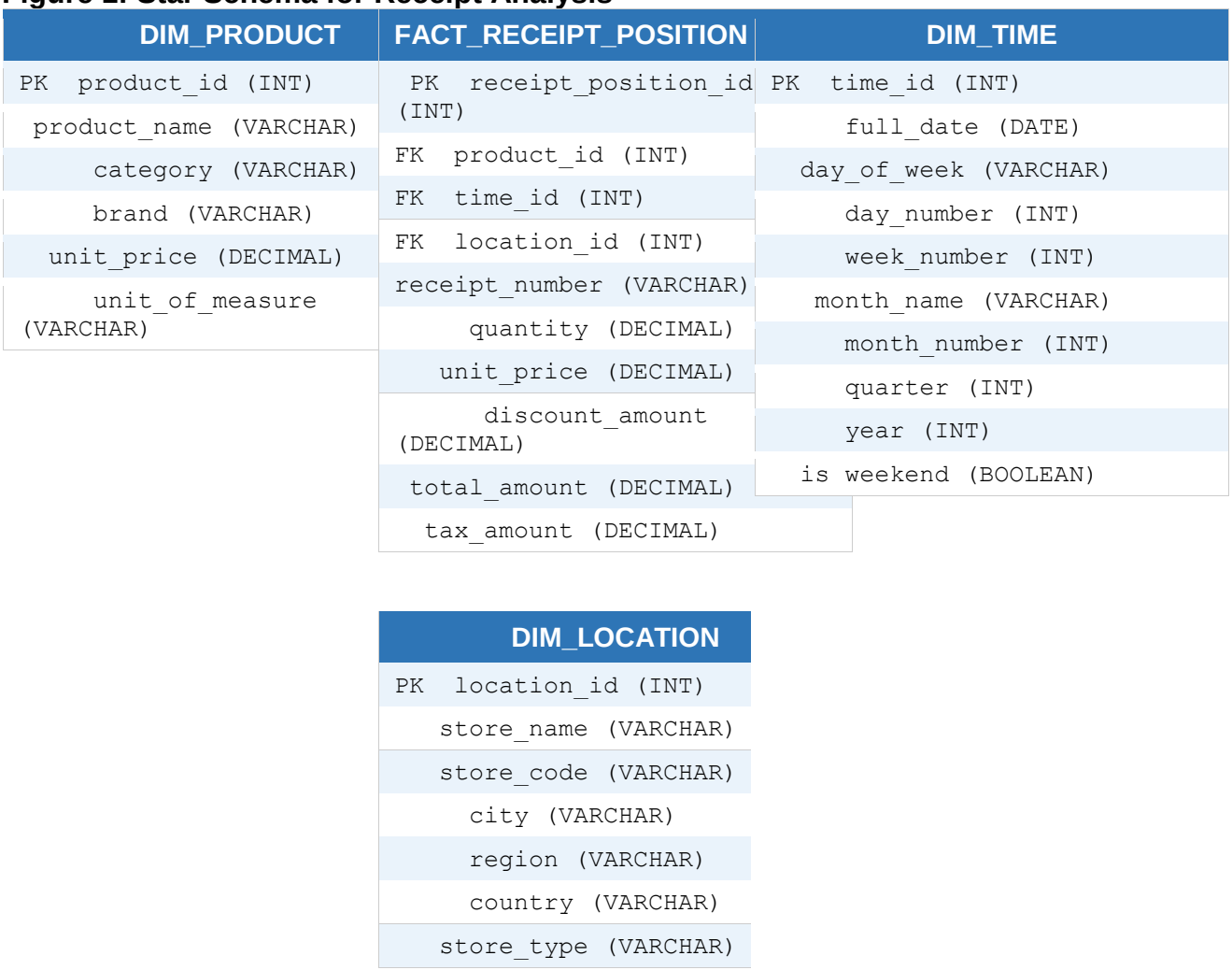
Sales volumes, quantities, prices, and other quantifiable, quantitative information regarding business events, often referred to as measures or metrics, are included in the fact table. The descriptive attributes, such as product names, dates, and store locations, that are used to filter, organize, and label the facts are contained in surrounding dimension tables. This design's main benefit is that it reduces the amount of joins needed in analytical queries, which makes it extremely effective for end users and reporting tools.

3.2 Receipt-Based Star Schema Design

Three dimensions, Product, Time, and Location, as well as one fundamental fact, are identified to represent the shopping receipt using the star schema. Each line item (receipt position) on the receipt itself represents a product that was bought in a specified quantity, at a certain price, at a specific time, and in a specific retail location.

The core of the model is the Fact table, FACT_RECEIPT_POSITION, which documents every single line item on the receipt. It uses foreign keys to relate to the three-dimensional tables. All descriptive characteristics pertinent to the analytical queries a company may pose are included in the dimension tables.

Figure 1: Star Schema for Receipt Analysis



The aforementioned star schema offers a clear and simple concept. Questions like "Which products generated the highest revenue last quarter?" can be answered by analysts. On weekends, which shop locations had the highest volume of transactions? These queries are quite effective in traditional BI systems since they only need one join between the fact data and the pertinent dimension table.
Product-level characteristics, including name, category, brand, and unit pricing, are recorded in the DIM_PRODUCT database.
Time-based aggregations at various granularities are made possible by DIM_TIME's extensive time dimension. Geographical and store-type analysis are supported by DIM_LOCATION. Quantities, prices, discounts, totals, and tax amounts are recorded as measurable facts in the fact table FACT_RECEIPT_POSITION for every line item on a receipt.

4. DATA VAULT DATA MODEL

4.1 Applying Data Vault to the Receipt

It takes a whole different perspective to apply the Data Vault technique to the identical receipt scenario. The Data Vault strategy optimises for data integration, auditability, and long-term scalability rather than query performance from the start. The concept separates descriptive context from bare identity and is organised around business keys and their connections.

The Receipt (transaction), the Product, the Time Period, and the Location are the four fundamental business entities for the receipt-based model. Every creature turns into a hub. A Link is used to depict the relationships between these entities that were recorded in the receipt line item event. Every descriptive property is transferred into satellite tables that are connected to the appropriate hub or link.

4.2 Hub Tables

The Data Vault model's solid foundation is made up of the four Hub tables. In order to maintain the model's fundamental identity even if source systems change, each Hub only keeps the business key for the corresponding entity.

Figure 2: Data Vault Hub Tables

HUB_RECEIPT	HUB_PRODUCT	HUB_TIME	HUB_LOCATION
PK receipt_hash_key	PK product_hash_key	PK time_hash_key	PK location_hash_key
receipt_number	product_code	full_date	store_code
load_date	load_date	load_date	load_date
record_source	record_source	record_source	record_source

4.3 Link Tables

The transactional relationship between the four Hubs is captured in the Link table LINK_RECEIPT_POSITION. A single receipt line item event is represented by each row in the Link, which links a particular receipt, product, time, and place. There are no descriptive qualities stored in the Link; just hash keys and metadata are.

Figure 3: Data Vault Link Table

LINK_RECEIPT_POSITION	
PK	receipt_position_hash_key
FK	receipt_hash_key
FK	product_hash_key
FK	time_hash_key
FK	location_hash_key
	load_date
	record_source

4.4 Satellite Tables

All descriptive and historical data are kept in satellite tables. Satellites are connected to each Hub and the Link. The load_date and load_end_date columns are used by the historization process to monitor changes over time, guaranteeing that no data is ever erased or replaced.

Figure 4: Data Vault Satellite Tables

SAT_PRODUCT	SAT_LOCATION	SAT_RECEIPT_POS	SAT_TIME
FK product_hash_key	FK location_hash_key	FK receipt_pos_hash_k	FK time_hash_key
load_date	load_date	load_date	load_date
load end date	load end date	load end date	load_end_date
hash_diff	hash_diff	hash_diff	hash_diff
product_name	store_name	quantity	day_of_week
category	city	unit_price	week_number
brand	region	discount amount	month_name
unit_price	country	total_amount	quarter
record source	store type	tax amount	year
	record_source	record_source	record_source

Separating context from identification is a crucial design idea that is shown by the satellite tables above. SAT_PRODUCT monitors changes in product data over time; if a product's category or price changes, a new record is created with a new load_date instead of replacing the old one. This keeps the entity's qualities intact over its entire history. SAT_LOCATION, SAT_TIME, and SAT_RECEIPT_POS may all be historised using the same method.
Each Satellite's hash_diff column contains a calculated hash of every descriptive property. This column is compared to incoming records throughout the ETL process to ascertain whether a change has taken place, allowing for effective delta detection without requiring a complete column-by-column comparison (Linstedt & Olschimke, 2015).

5. MODELING TECHNIQUES ADVANTAGES AND DISADVANTAGES

5.1 Advantages and Disadvantages of Star Schema

The star schema has been the dominant data warehouse modeling technique for over two decades, and its widespread adoption reflects a number of genuine strengths. However, it also has structural limitations that become evident in complex or rapidly evolving environments.

Advantages

- **Simplicity and Intuitiveness**: Both developers and business users find the star schema to be quite straightforward. It is simple to comprehend and navigate since the division of data and dimensions readily translates into business principles.
- **Query Performance**: Queries usually need fewer joins since the schema is deformalized. This leads to quicker query execution times, which are especially advantageous for reporting tools and large-scale analytical workloads (Corr & Stagnitto, 2011).
- **Compatibility with BI technologies**: Star schema topologies are optimised for the majority of contemporary BI and reporting technologies, including Power BI, Tableau, and QlikView. These technologies' semantic models and metadata layers fit well with fact-dimension structures.
- **Implementation Ease:** The star schema may be swiftly implemented and maintained with very little overhead in stable systems with clear needs.

Disadvantages

- **Limited Flexibility:** When new source systems, data entities, or business needs arise, it is challenging to expand the star schema. Significant redesign is frequently

necessary for structural changes, including adjustments to fact tables and ETL pipelines.

- **Data Redundancy:** Dimension tables experience data redundancy as a result of denormalisation. The possibility of storing the same descriptive data more than once raises storage needs and the possibility of inconsistent data.
- **Inadequate Auditability:** Full historization is not natively supported by the star schema. Although they can be used, Slowly Changing Dimensions (SCDs) are more complicated and do not offer the same degree of audit trail as the Data Vault (Corr & Stagnitto, 2011).
- **Stability Assumption:** The design is predicated on the relative stability of the source systems and underlying business processes. This assumption often fails in situations that are nimble or changing quickly.

5.2 Advantages and Disadvantages of Data Vault

Advantages

- **Scalability and Flexibility**: New Hubs, Links, and Satellites may be added to the Data Vault without changing the current tables because to its modular design. Because of this, it works incredibly well in agile development settings where needs are constantly changing (Collier, 2011).
- **Complete Auditability and Historical Monitoring**: Each entry in the Data Vault has a source and load date. Satellites record all changes in descriptive properties throughout time. Because of this, the Data Vault is automatically compliant with audit and data governance regulations.
- **Parallel Loading**: All tables may be loaded in parallel as Hubs, Links, and Satellites are independent of one another. For big data quantities, this greatly enhances ETL performance (Linstedt & Olschimke, 2015).
- **Resilience to Source System Changes**: The model is unaffected by the replacement or restructuring of source systems since Hubs are built on business keys rather than system-generated surrogate keys.
- **Support for numerous Sources**: Integrating data from numerous source systems without conflict or ambiguity is made simple by the record_source field in each table.

Disadvantages

- **Complexity:** Compared to the star schema, the Data Vault concept is far more complicated. The costs associated with creating and maintaining Hubs, Links, and Satellites may exceed the advantages for small teams or projects with straightforward, steady requirements.
- **Query Complexity:** Several joins across Hubs, Links, and Satellites are necessary to query the raw Data Vault directly. To make data available for end-user reporting, an extra transformation layer (such as an Information Mart or Business Vault) is usually needed (Hultgren, 2012).
- **Longer Initial Development Time:** Compared to the star schema, developing a Data Vault architecture necessitates more upfront design work and a greater degree of technical skill.
- **Performance Overhead**: If proper optimisation structures like PIT and Bridge tables are not added, the normalised structure and historization method may cause performance issues for analytical workloads that involve a lot of reading.

5.3 Comparative Summary

The two modelling techniques are compared side by side across important business data warehousing parameters in the table that follows.

Dimension	Star Schema	Data Vault
Structure	Denormalized (fact + dimensions)	Highly normalized (Hub, Link, Satellite)
Query Performance	High (few joins required)	Moderate (requires transformation layer)
Flexibility	Low (structural changes are costly)	High (add tables without changing existing ones)
Auditability	Limited (requires SCD implementation)	Full (built-in historization and record source)
ETL Complexity	Moderate	Higher (parallel load patterns required)
Agile Suitability	Low to Moderate	High
BI Tool Compatibility	Very High	Moderate (requires Information Mart layer)
Best Use Case	Stable, well-defined reporting needs	Complex, evolving, multi-source environments

6. CONCLUSION

The star schema and the Data Vault method, two popular data modelling techniques for business intelligence and data warehousing, have been thoroughly examined in this project report using the real-world example of a retail shopping receipt.

For many conventional BI applications, the star schema is still the best option because of its fact-and-dimension design, which provides an easy-to-use and query-optimized structure. It is quite useful in stable, well-defined contexts because of its ease of use and interoperability with common reporting tools. However, when new data sources need to be connected or business objectives alter, its rigidity becomes a problem.

In contrast, the Data Vault approach offers a solid basis for long-term, flexible data warehouse expansion. Modular, simultaneous, and auditable data loading is made possible by its division of business keys (Hubs), relationships (Links), and descriptive qualities (Satellites). With hash key optimization, Big Data compatibility, and a well-defined layered architecture, Data Vault 2.0 expands these capabilities even further. This report's receipt-based Data Vault model shows how even a straightforward transactional dataset may be modelled to provide full historical traceability and future scalability.

According to the comparison study, neither strategy is always better. The stability of source systems, the rate of business change, the size of the development team, and the type of analytical needs all influence the right decision. The Data Vault approach has great long-term benefits for businesses working in dynamic, data-intensive settings with stringent governance and audit needs. The star schema is still a great and practical option for businesses with more straightforward and dependable BI requirements.

In reality, a lot of contemporary data warehouse designs take a hybrid approach, storing raw data in the integration layer using Data Vault and reporting utilizing star schema structures in

the consumption layer. This combination is an emerging best practice in business data architecture, utilizing the advantages of both approaches.

7. PROCESS REFLECTION AND DOCUMENTATION

The evaluation criteria outlined in the IU project report requirements are covered in this chapter, which also documents the project's methodology, design thinking techniques used, resource management, and a critical evaluation of the caliber and originality of the results.

7.1 Transfer: Application of Design Thinking

Design thinking is an iterative, human-centered approach to problem-solving that breaks down difficult problems into smaller, more achievable steps. The design thinking methodology was modified and used in five stages for this project report: Empathies, Define, Ideate, Prototype, and Test.

The problem context was thoroughly examined throughout the Empathise phase. The assignment requires a practical demonstration of the Data Vault approach's applicability to a real-world situation in addition to a theoretical comprehension of the method. Because the shopping receipt is a globally accessible business artefact that encapsulates the fundamental complexity of transactional data, multiple entities (product, time, place), measurable facts (quantity, price, totals), and connections between them, it was selected as the use case. This option demonstrated a sympathetic awareness of the target audience, which consists of assessors who must comprehend the rationale behind modelling choices.

The report's scope was explicitly defined throughout the Define phase. Task 2 from the assignment brief, comparing the Data Vault and Star Schema approaches, was the main focus. How can a retail receipt be modelled in a data warehouse in a way that enables both historical traceability and analytical flexibility? This was the main issue that needed to be resolved.

The two modelling techniques were investigated concurrently throughout the Ideate phase. The Data Vault technique was seen as the more advanced option, and the star schema was initially taken into consideration as the established baseline. A like-for-like comparison was made possible by mapping both methods to the same source data (the receipt). Before being formalised into the structured table diagrams shown in Sections 3 and 4, preliminary drawings of both data models were created.

Converting the conceptual data models into tangible table designs with column names, data types, and keys was the task of the prototype phase. This process necessitated several iterations, especially for the Data Vault model, where choices had to be made about the granularity of Satellites and the proper distribution of characteristics among Hub-level and Link-level Satellites.

During the test phase, each model was compared to the business questions that a retail analyst may have, like: Which goods sold the most in a certain month? Which retail locations brought in the most money? Despite having varying degrees of complexity and querying strategies, this validation stage verified that both models could satisfy the desired analytical use cases.

Given the assignment's clear definition, one major change to the typical design thinking method was to combine the Empathise and Define stages into a single scoping activity. This was a purposeful efficiency choice because the assignment requirement itself included enough background information to get straight into modelling without doing a lot of user research. This modification is in line with agile project concepts, which place an emphasis on functional results rather than copious documentation (Collier, 2011).

7.2 Process Documentation

This project report was developed using an organised, iterative method. The project was divided into four primary phases, each of which built upon the results of the one before it.

Stage 1: Literature Review and Research: In the first stage, the required literature was thoroughly reviewed, with a focus on Linstedt and Olschimke (2015), Hultgren (2012), and Collier (2011). Prior to attempting to implement the Data Vault approach in practice, the objective was to provide a strong theoretical foundation for it. To ensure that contemporary industry viewpoints were covered and to augment the required readings, additional academic and practitioner sources were explored.

Stage 2: Conceptual Modelling: After establishing the theoretical framework, the receipt scenario was examined to determine the primary business entities, their connections, and the quantifiable details connected to every transaction. Two conceptual models were drawn, one using the Data Vault technique and the other using the star schema approach. In order to provide a fair comparison, special care was given at this point to making sure that both models represented the same underlying data.

Stage 3: Detailed Model Design: Using table names, column definitions, primary and foreign keys, and data types, the conceptual models were developed into logical data models. In compliance with Data Vault 2.0 requirements, each table was updated with hash key columns, load dates, finish dates, hash difference columns, and record source columns for the Data Vault model (Linstedt & Olschimke, 2015).

Stage 4: Writing and Synthesis: In this last phase, the research findings, model diagrams, and comparative analysis were integrated into a cohesive narrative while the report was written in an organised academic fashion. Several improvements to the models were made as a result of the writing process itself, as the process of elucidating design choices highlighted areas that need further clarification. During this phase, for instance, the designs of the satellite tables were updated to more accurately reflect the source-separation principle, which states that characteristics from various source systems should preferably be kept in distinct satellites.
Continuous reflection was maintained throughout the procedure. The results were checked for alignment with the assignment requirements after each step. By using an iterative self-review process, holes were found early on and significant rework was prevented later.

7.3 Resources

The required course reading, publicly accessible scholarly and professional resources, and the IU assignment brief itself were among the resources available for this research. In keeping with the parameters of a mid-complexity project report, time was handled over the course of around two weeks.

All of the report's technical statements were primarily supported by the recommended literature. The canonical source for Data Vault 2.0 specs and nomenclature was Linstedt and Olschimke (2015). The dimensional modelling methods used in Section 3 and the construction of the star schema were inspired by Corr (2011). Additional viewpoints on rapid data warehouse development and the real-world implementation of Data Vault were offered by Collier (2011) and Hultgren (2012).

About 25% of the time was devoted to research and literature study, 35% to data model creation and diagramming, and 40% to writing, revising, and improving the report. The modeling-intensive nature of the endeavor, which necessitated meticulous and iterative design effort to develop precise and well-structured diagrams, is reflected in this distribution. Other than conventional academic materials, no other tools were needed. The data models were created theoretically and recorded in the report's structured table format. Since the assignment concentrated on logical data modelling rather than actual database building, no software implementation was done.

7.4 Creativity and Innovation

Similar examples may be found in the literature and online educational resources, and data modelling reports of this kind are a well-established genre in academic and professional settings. This report's contribution is found in the depth and clarity with which it applies current techniques to a specific, real-world situation rather than in the presentation of wholly original ideas.

The retail purchase receipt is used as the modelling scenario, which offers a realistic and approachable starting point for comprehending the Data Vault and star schema techniques. Although data warehousing textbooks frequently use receipts as an example, this report's particular methodology—building both models in parallel from the same source artefact and then directly comparing them side by side—offers a more comprehensive and educational analysis than is usually found in introductory treatments of the subject.

The clear mapping of Data Vault 2.0 standards to the receipt model, which includes the usage of hash keys, hash difference columns, and load finish dates in the Satellite tables, is another creative aspect of the report. Many Data Vault examples seen in textbooks stop at the conceptual level and exclude these technological specifics. By incorporating them, this study provides practitioners with tangible benefits by bridging the gap between theory and practice.
Furthermore, the comparative summary table in Section 5.3 summarises the benefits and drawbacks of both strategies across eight analytical dimensions, offering an organised framework for making decisions that goes beyond the typical qualitative discussion seen in the majority of studies that are similar.

7.5 Quality Assessment

The assignment's main issue was how to use the Data Vault approach in a situation that relies on receipts and how it differs from the star schema. That issue is immediately and thoroughly addressed in this study. The data models created in Sections 3 and 4 are based on the required literature, are technically correct, and make sense. Section 5's comparative analysis offers a fair and fact-based assessment of both strategies.

The three focal areas listed in the assignment brief can be used to gauge the quality of the outcome. First, using four of the recommended sources, Section 2's study of the Data Vault approach discusses its history, essential elements, and improvements made in version 2.0. Second, the data models in Sections 3 and 4 appropriately and technically implement the related Hub, Link, and Satellite structure in the Data Vault, as well as the necessary one fact table and three dimension tables in the star schema. Third, the benefits and drawbacks discussed in Section 5 are precise, well-balanced, and backed up by references.

The report's presentation of the data models as logical ideas rather than actual implementations is one of its limitations. Although they were outside the purview of this project, physical factors like indexing techniques, partitioning, and query optimization would be crucial in a real-world implementation. Similarly, since loading data into the Data Vault model was not a necessary deliverable for Task 2, the ETL procedure was not described. These fields are logical extensions of the work that might be investigated in more research. All things considered, the paper effectively accomplishes the goals outlined in the project brief and offers a comprehensive, organized, and realistically grounded study of the Data Vault technique and its comparison with the star schema.

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